



CSCE 633-700: Machine Learning

Support Vector Machines

Overview

- Lagrange Optimization
- Support Vector Machines
 - Linearly separable case
 - Non-separable case
 - Hinge Loss
 - Multi-class SVMs

Overview

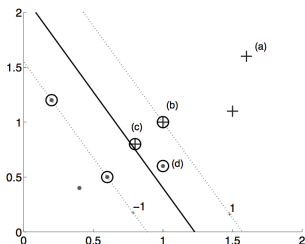
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Support Vector Machines: Non-separable case

- If two classes are not linearly separable, we look for the hyperplane that yields the least error
- We define **slack variables** $\xi_n \geq 0$ which represent the deviation from the margin

$$y_n(\mathbf{w}^T \mathbf{x}_n + w_0) \geq 1 - \xi_n$$

- *Case (a)*: Far away from the margin, $\xi_n = 0$
- *Case (b)*: On the right side and on the margin, $\xi_n = 0$
- *Case (c)*: On the right side, but in the margin, $0 \leq \xi_n \leq 1$
- *Case (d)*: On the wrong side, $\xi_n \geq 1$



We incorporate the number of misclassifications $\#\{\xi_n > 1\}$ and the number of non-separable points $\#\{\xi_n > 0\}$ as a **soft error** $\sum_n \xi_n$.

Support Vector Machines: Non-separable case

$$\min_{\mathbf{w}, \xi} \left[\frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{n=1}^N \xi_n \right], \quad \text{s.t. } y_n(\mathbf{w}^T \mathbf{x}_n + w_0) \geq 1 - \xi_n \text{ and } \xi_n \geq 0, \quad \forall n$$

1) Formulate Lagrangian function (primal problem)

$$L_p = \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{n=1}^N \xi_n - \sum_{n=1}^N \alpha_n [y_n(\mathbf{w}^T \mathbf{x}_n + w_0) - 1 + \xi_n] - \sum_{n=1}^N \mu_n \xi_n$$

2) Minimize Lagrangian to solve for primal variables $\mathbf{w}$, w_0

$$\frac{\partial L_p}{\partial \mathbf{w}} = 0 \Rightarrow \mathbf{w} = \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n$$

$$\frac{\partial L_p}{\partial w_0} = 0 \Rightarrow \sum_{n=1}^N \alpha_n y_n = 0$$

$$\frac{\partial L_p}{\partial \xi_n} = 0 \Rightarrow C - \alpha_n - \mu_n = 0$$

Support Vector Machines: Non-separable case

3) Substitute the primal variables $\mathbf{w}$, w_0 into the Lagrangian and express in terms of dual variables α_n

$$\begin{aligned}
 L_d &= \frac{1}{2} \|\mathbf{w}\|_2^2 + C \underbrace{\sum_{n=1}^N \xi_n}_{\text{slack variables}} - \mathbf{w}^T \sum_{n=1}^N \alpha_n y_n \mathbf{x}_n \\
 &\quad - w_0 \sum_{n=1}^N \alpha_n y_n + \sum_{n=1}^N \alpha_n - \underbrace{\sum_{n=1}^N \alpha_n \xi_n}_{\text{slack penalty}} - \underbrace{\sum_{n=1}^N \mu_n \xi_n}_{\text{slack penalty}} \\
 &= -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \sum_{n=1}^N \xi_n (C - \alpha_n - \mu_n) + \sum_{n=1}^N \alpha_n \\
 &= \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m
 \end{aligned}$$

Support Vector Machines: Non-separable case

4) Maximize the Lagrangian with respect to dual variables (dual problem)

$$\max_{\alpha_n} \left\{ \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m \right\}$$

s.t. $\sum_{n=1}^N \alpha_n y_n = 0$ and $0 \leq \alpha_n \leq C$, for $n = 1, \dots, N$

- Solved numerically using quadratic optimization methods
- $\alpha_n = 0$: instances at the correct side of the hyperplane with sufficient margin
- $\alpha_n > 0$: support vectors
 - $0 < \alpha_n < C$: instances lying on the margin
 - $\alpha_n = C$: instances in the margin or misclassified